

Session-12

1. Day of the Year:

```
class Solution {
    public int dayOfYear(String date) {
        int year = Integer.parseInt(date.substring(0, 4));
        int month = Integer.parseInt(date.substring(5, 7));
        int day = Integer.parseInt(date.substring(8, 10));

        int[] daysInMonth = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

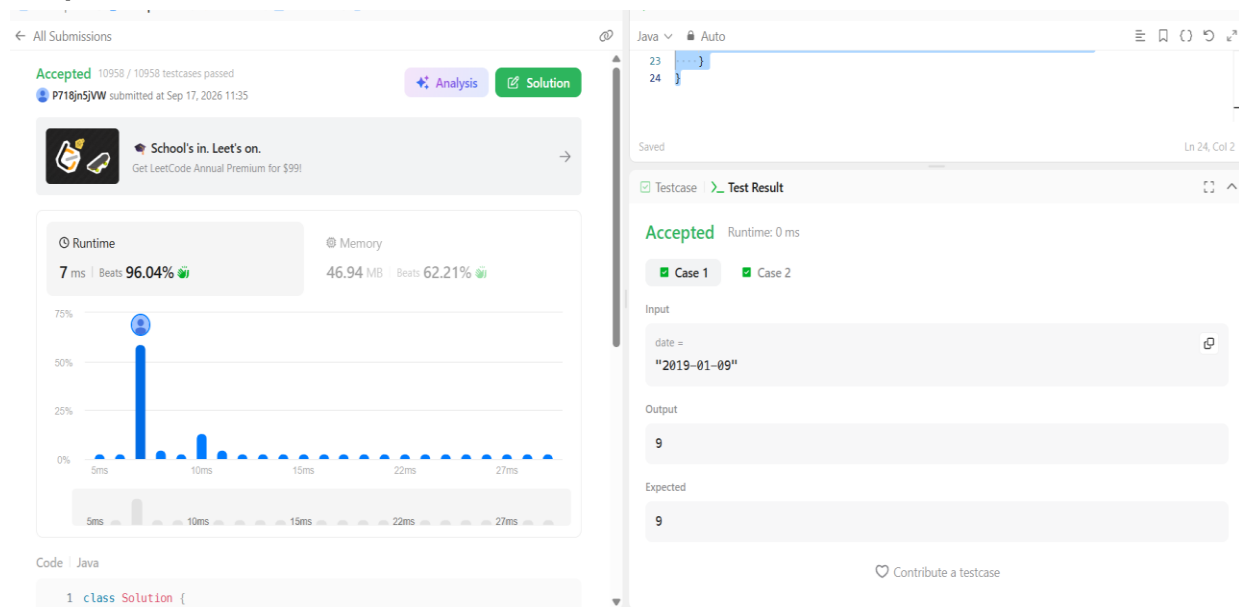
        if (isLeapYear(year)) {
            daysInMonth[1] = 29;
        }

        int dayOfYear = day;
        for (int i = 0; i < month - 1; i++) {
            dayOfYear += daysInMonth[i];
        }

        return dayOfYear;
    }

    private boolean isLeapYear(int year) {
        return (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
    }
}
```

Output:



2. Day of the Week:

```
class Solution {
    public String dayOfTheWeek(int day, int month, int year) {
        String[] days = {"Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday",
"Saturday"};
        int[] daysInMonth = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

        int totalDays = day;
```

```

for (int y = 1971; y < year; y++) {
    if (isLeapYear(y)) {
        totalDays += 366;
    } else {
        totalDays += 365;
    }
}

for (int m = 1; m < month; m++) {
    if (m == 2 && isLeapYear(year)) {
        totalDays += 29;
    } else {
        totalDays += daysInMonth[m - 1];
    }
}

return days[(totalDays + 4) % 7];
}

private boolean isLeapYear(int year) {
    return (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
}
}

```

Output:

The screenshot displays a LeetCode submission interface. On the left, the 'All Submissions' tab is active, showing an 'Accepted' status for 45/45 testcases. The submission was made by user 'P718jn5VW' on Sep 17, 2026 at 11:37. A performance bar chart shows the user's runtime (0 ms) is significantly faster than other submissions, with a 'Beats 100.00%' indicator. The memory usage is 42.36 MB, and the solution beats 84.30% of other submissions. On the right, the 'Test Result' section shows three test cases passed. The input for Case 1 is 'day = 31', 'month = 8', and 'year = 2019'. The output is 'Saturday'.